



Lab Number: 01

Course Code: CL2001

Course Title: Data Structures - Lab

Semester: Fall 2026

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Guidelines

- You may use **only the concepts covered in the lab** to complete this task.
- Your submission must be a **single PDF file** that includes:
 - The source code (in plain text)
 - Screenshots of the corresponding outputs
- Name your file using the following format: **rollno_name_labtasknumber.pdf**
Example: *25p-1234_ozair_labtask01.pdf*
- Unethical use of AI tools will result in **0 marks**.
- Late submissions will incur a penalty of **-3 marks per day**.

Tasks

ONLY USE POINTER NOTATION TO IMPLEMENT ALL THE SOLUTIONS. ARRAY INDEXING [] IS NOT ALLOWED.

1 Remove Duplicates In-Place

Given an array of n integers, remove all duplicate values without creating another array. The relative order of the remaining elements must be preserved.

Example:

Input: 4 2 4 7 2 9 4
Output: 4 2 7 9

The program should modify the original array and return the new logical size.

Required function:

```
int removeDuplicates(int* arr, int n);
```

2 Second Smallest Without Sorting

Find the second smallest distinct element of a given array.

Input: 10 8 5 10 3 5
Output: 5



If there is no second distinct value, report that.

Restriction: Do not sort the array.

Required: Solve using a single traversal.

3 Rearrange Positive and Negative Values

Given an array containing positive and negative integers, create a function that rearranges its elements so that negative values appear before positive values.

The relative ordering inside each group does not matter.

Example:

Input: 4 -2 7 -8 3 -1 5

Output: -2 -8 -1 4 7 3 5

Implement the solution without creating another array.

Required function:

```
void rearrangeArray(int* arr, int size);
```